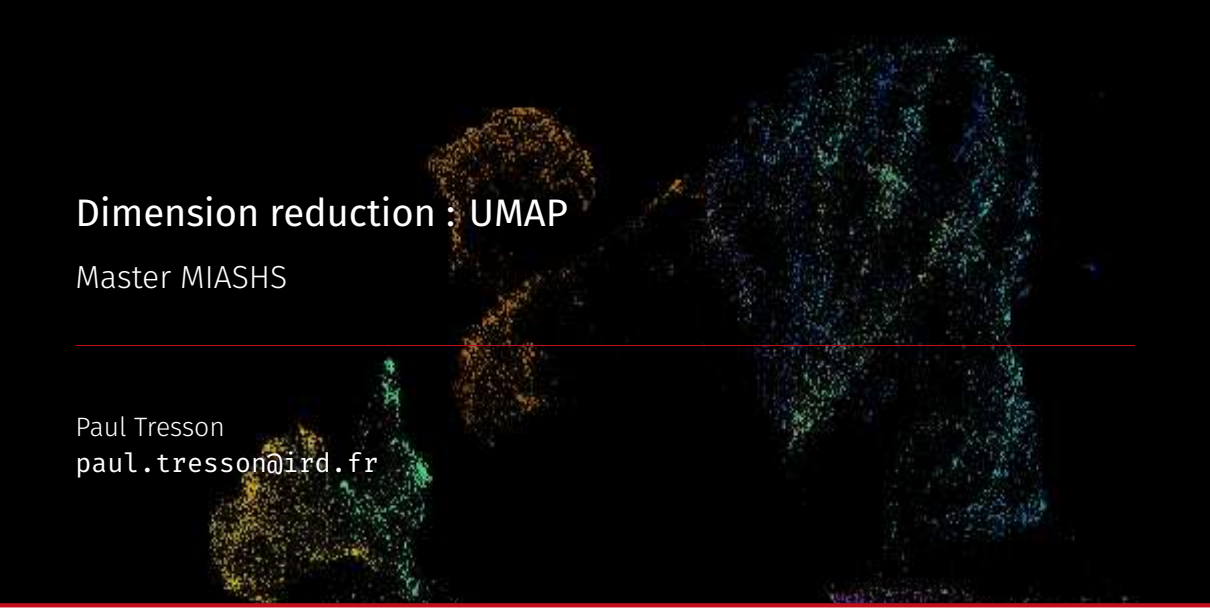


# Dimension reduction : UMAP

Master MIASHS

Paul Tresson
paul.tresson@ird.fr

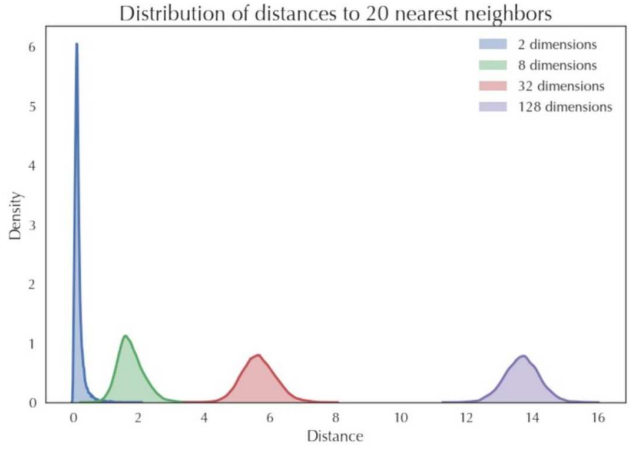
Institut de Recherche pour le Développement FRANCE
AMAPlab
UNIVERSITÉ PAUL VALÉRY MONTPELLIER

# Why reduce dimensions ?

# Why reduce dimensions ?

- Visualization and exploration
- Dimensions can be redundant
- Dimensions can be non-informative
- **Curse of dimensionality**

# Curse of dimensionality



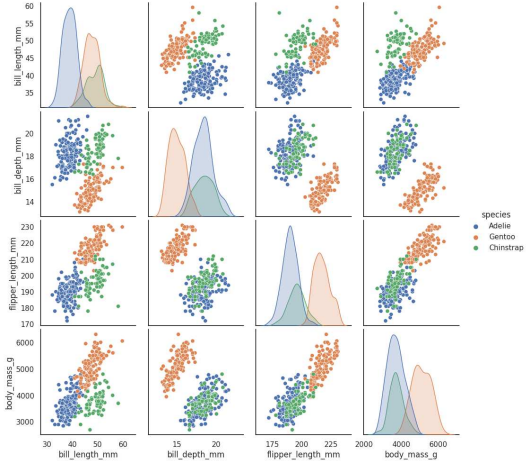

Leland McInnes slides

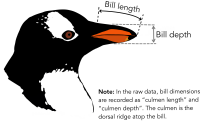
# Working with less dimensions can help

- Clustering
- Outlier detection
- Scaling up
- ...

# Visualize what is happening in higher dimensions



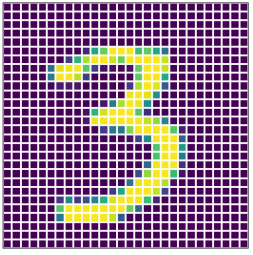



# Visualize what is happening in higher dimensions

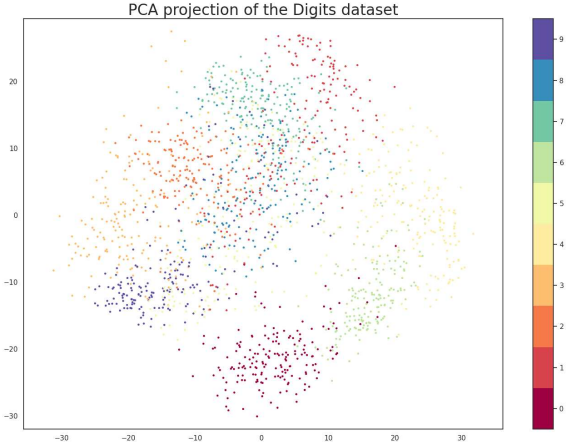


# Visualize what is happening in higher dimensions

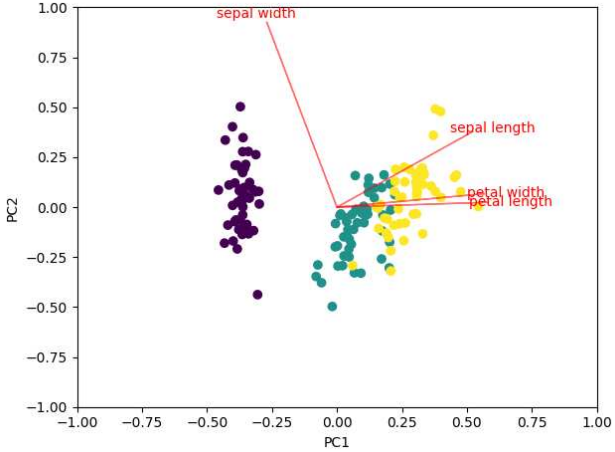


| label | pixel 001 | ... | pixel 350 | pixel 351 | pixel 352 | ... | pixel 728 |
|---|---|---|---|---|---|---|---|
| 3 | 0 | | 253 | 253 | 187 | | 0 |
| 1 | 0 | | 228 | 245 | 51 | | 0 |
| ⋮ | ⋮ | | ⋮ | ⋮ | ⋮ | | ⋮ |

# PCA


PCA projection of the Digits dataset
20
10
0
−10
−20
−30
−30
−20
−10
0
10
20
30
9
8
7
6
5
4
3
2
1
0

# PCA interpretability



# PCA advantages and drawbacks

## Advantages

- fast

## Drawbacks

# PCA advantages and drawbacks

## Advantages

- fast
- scales well

## Drawbacks

# PCA advantages and drawbacks

## Advantages

- fast
- scales well
- explainatory

## Drawbacks

# PCA advantages and drawbacks

## Advantages

- fast
- scales well
- explainatory

## Drawbacks

- not suited for non-linear data

# Non linear data

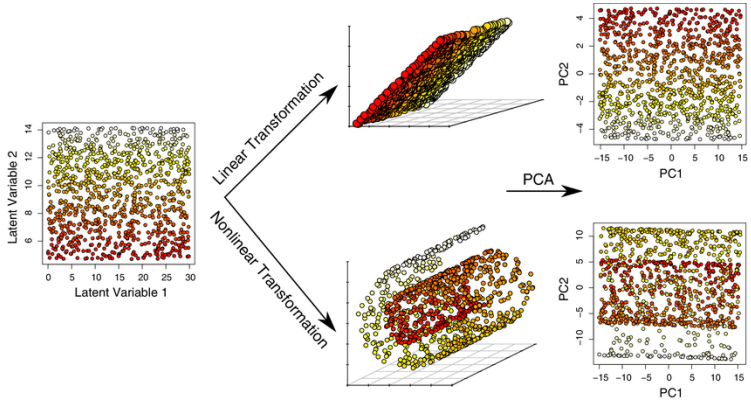
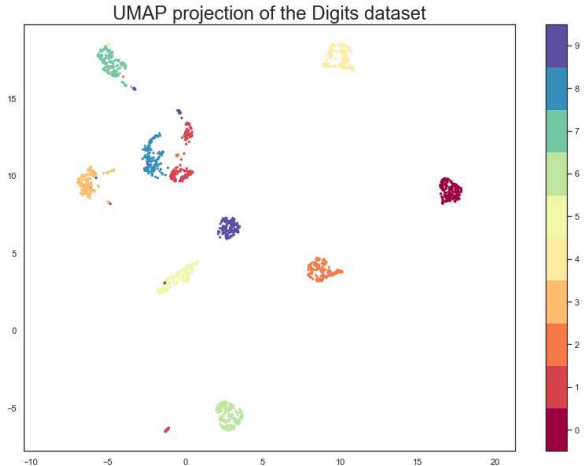



Figure from Du, 2019

# What is a manifold ?

In mathematics, a **manifold** is a topological space that locally resembles Euclidean space near each point. More precisely, an $n$-dimensional manifold, or *$n$-manifold* for short, is a topological space with the property that each point has a neighborhood that is homeomorphic to an open subset of $n$-dimensional Euclidean space.

# Non linear data



# How to work with non linear data

1. Find a good representation of the data in high dimension
2. Fit a good representation of in low dimension

# Classics

- **SNE** (Stochastic Neighbor Embedding) Hinton and Roweis, 2002

# Classics

- **SNE** (Stochastic Neighbor Embedding) Hinton and Roweis, 2002
- **T-SNE** (T-distributed SNE) Van der Maaten and Hinton, 2008

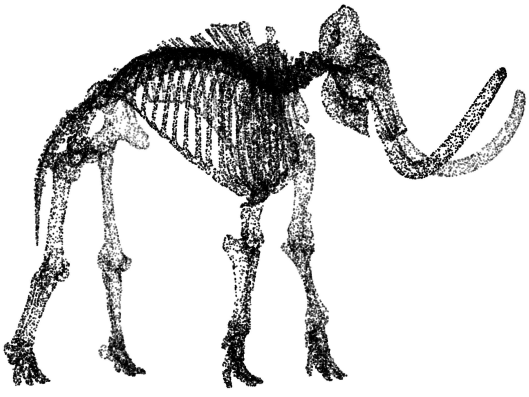
# Classics

- **SNE** (Stochastic Neighbor Embedding) Hinton and Roweis, 2002
- **T-SNE** (T-distributed SNE) Van der Maaten and Hinton, 2008

- **UMAP** (Uniform Manifold Approximation and Projection) McInnes et al., 2018

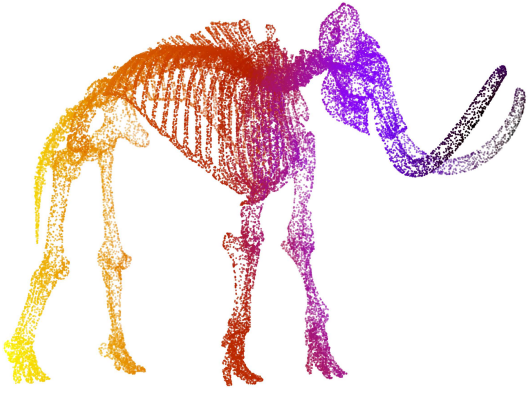
# How does UMAP work ?

# Mammoth example



Mammoth dataset by the Smithonian

# Mammoth example

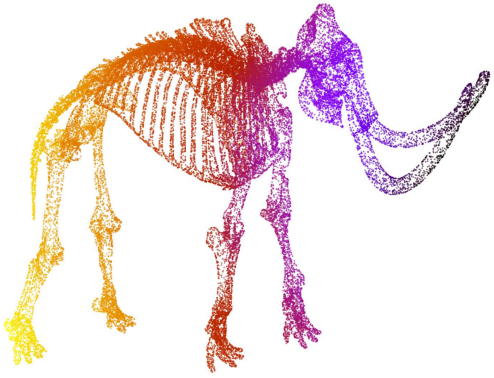


Mammoth dataset by the Smithonian

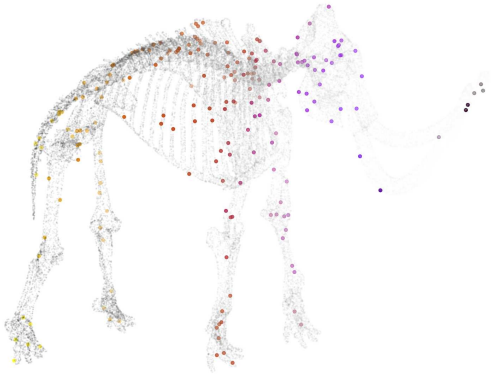
# PCA



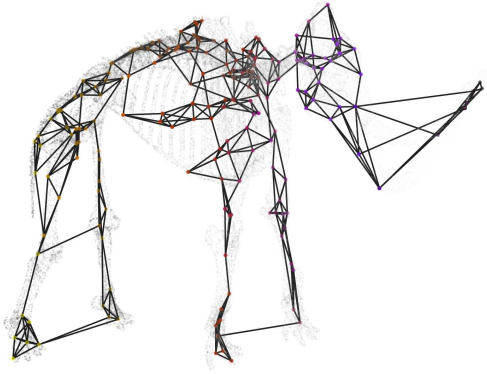
# Finding high dimension graph



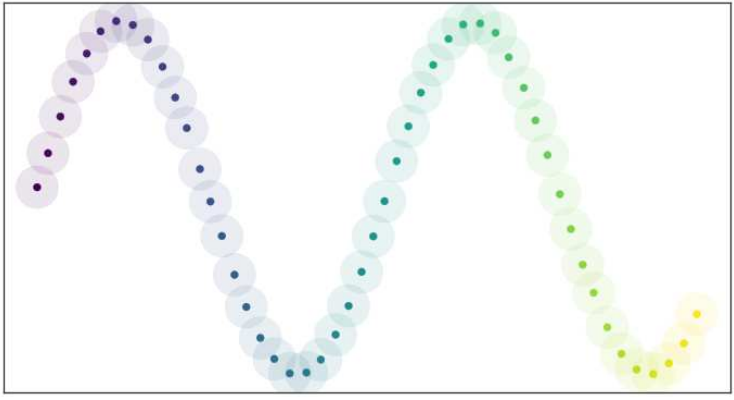
# Finding high dimension graph



# Finding high dimension graph

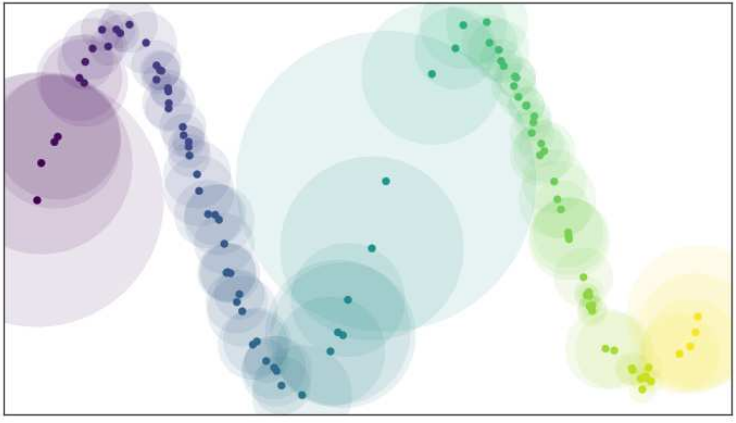


# *Uniform* Manifold Approximation and Projection



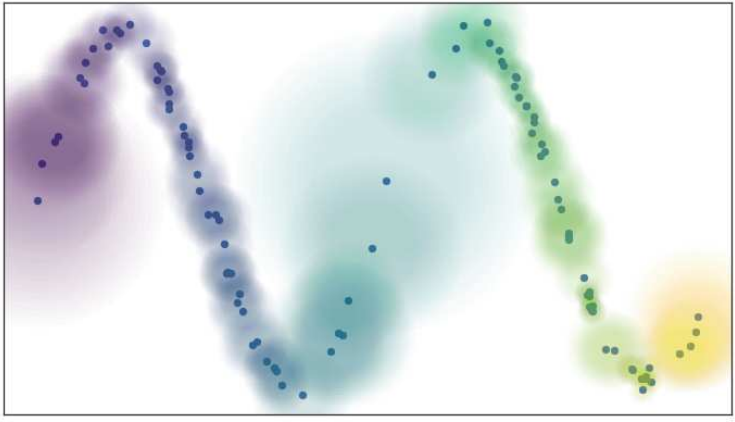
UMAP documentation

# *Uniform* Manifold Approximation and Projection



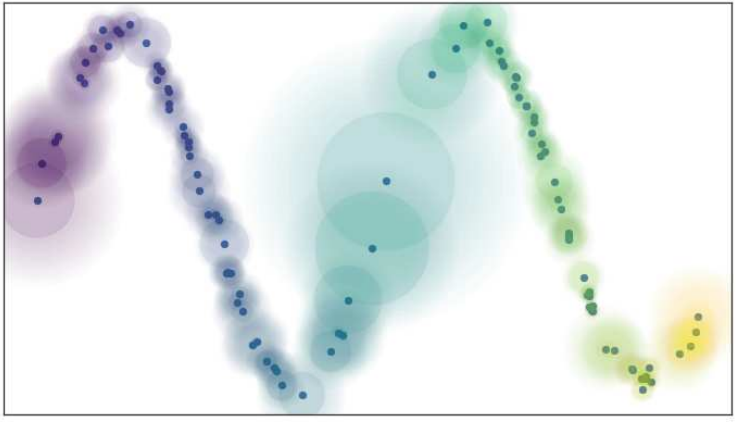
UMAP documentation

# *Uniform* Manifold Approximation and Projection



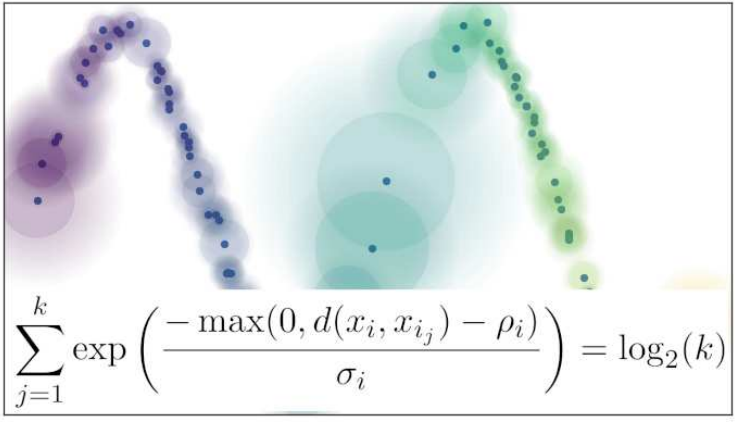
UMAP documentation

# *Uniform* Manifold Approximation and Projection



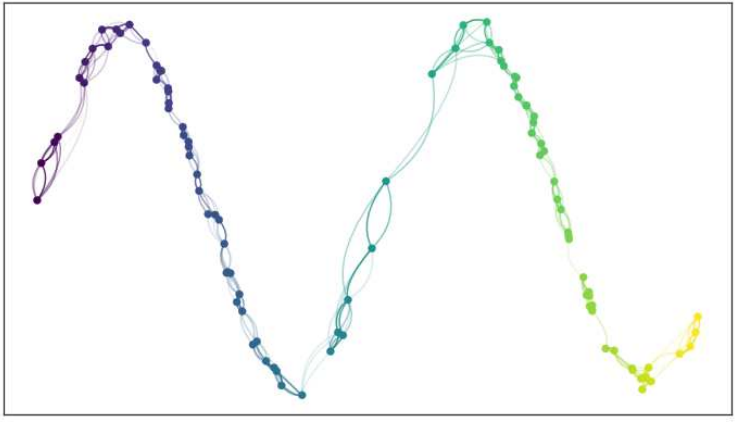
UMAP documentation

# *Uniform* Manifold Approximation and Projection



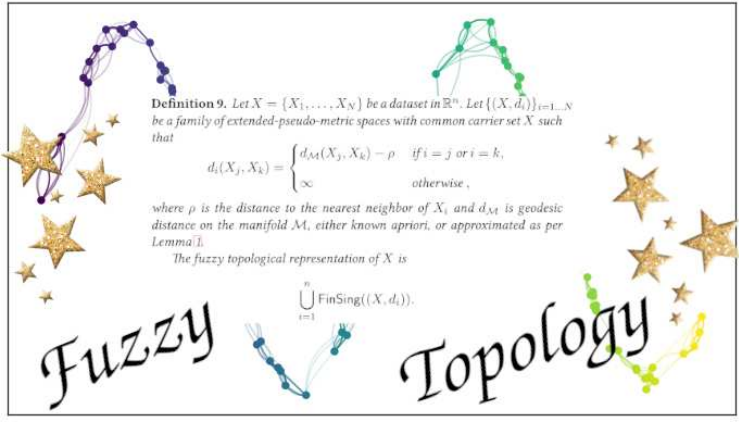

UMAP documentation

# *Uniform* Manifold Approximation and Projection



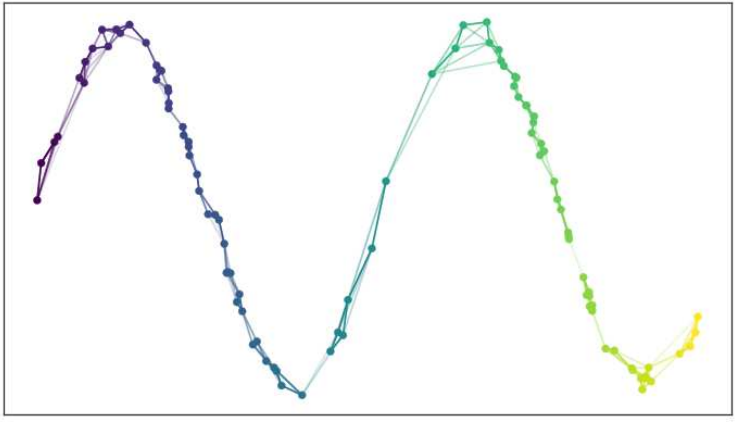
UMAP documentation

# *Uniform* Manifold Approximation and Projection



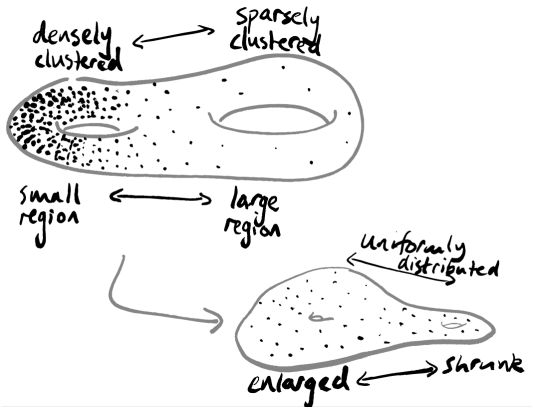

UMAP documentation

# *Uniform* Manifold Approximation and Projection



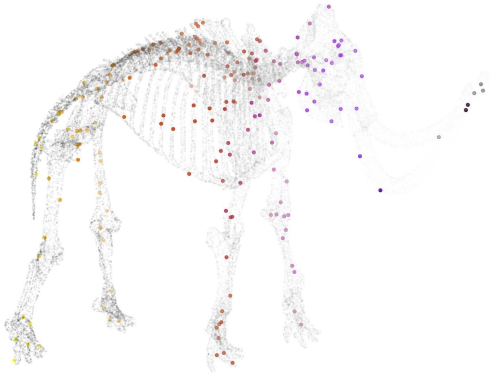
UMAP documentation

# *Uniform* Manifold Approximation and Projection

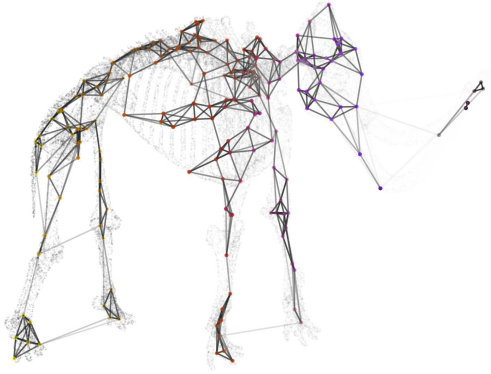



Tim Hosgood - Understanding UMAP

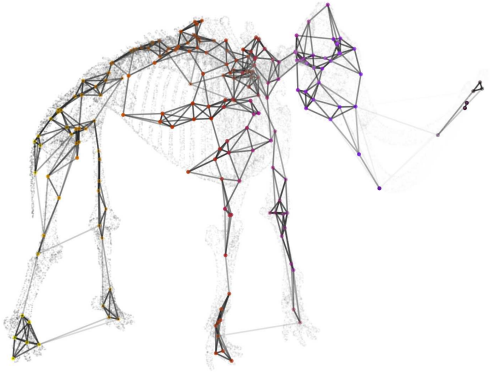
# Finding high dimension graph



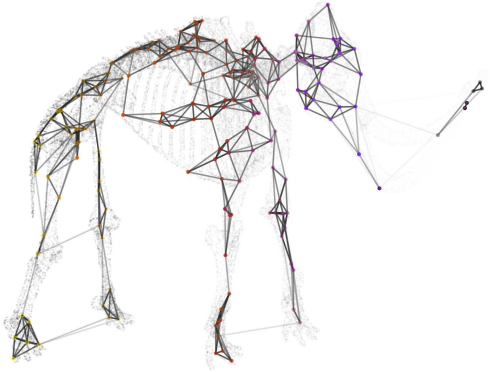
# Finding high dimension graph

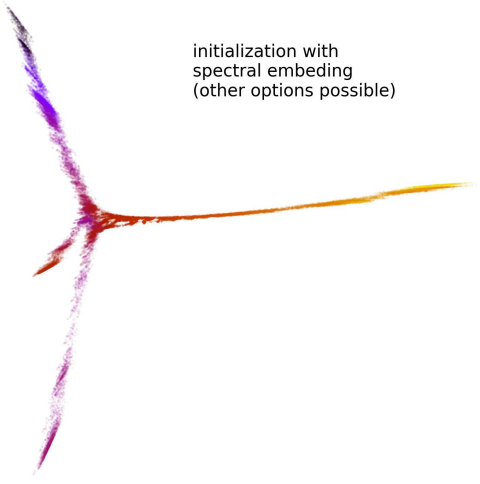


# Fitting low dimensional representation



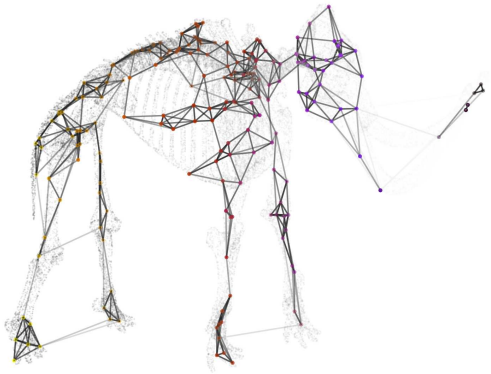
# Fitting low dimensional representation

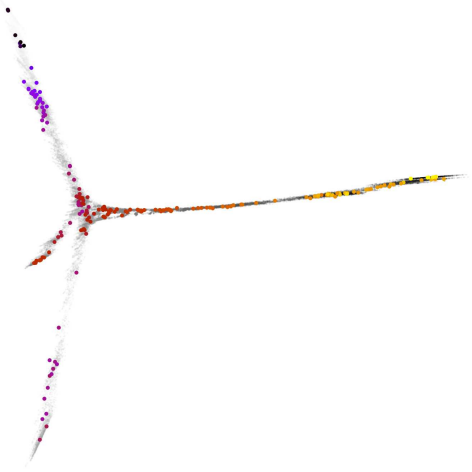




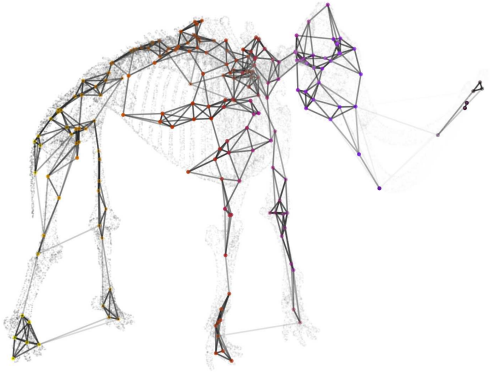
initialization with
spectral embeding
(other options possible)

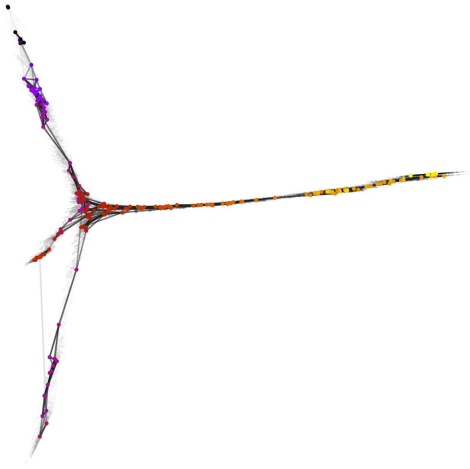
# Fitting low dimensional representation



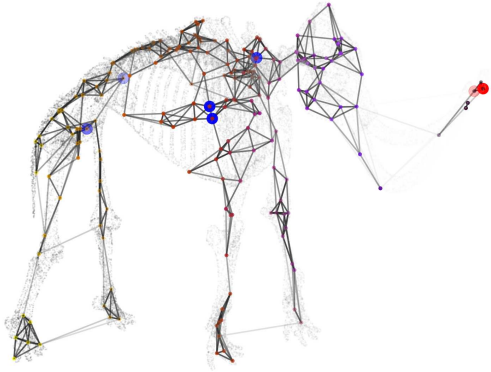


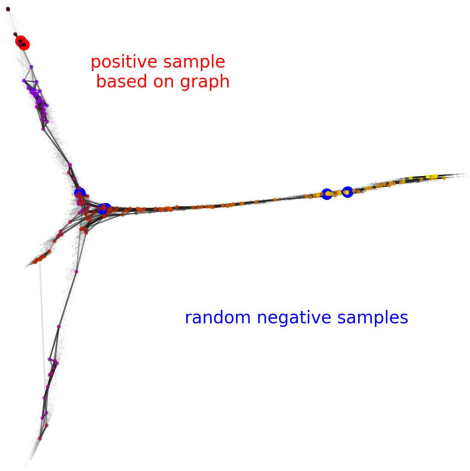
# Fitting low dimensional representation





# Fitting low dimensional representation





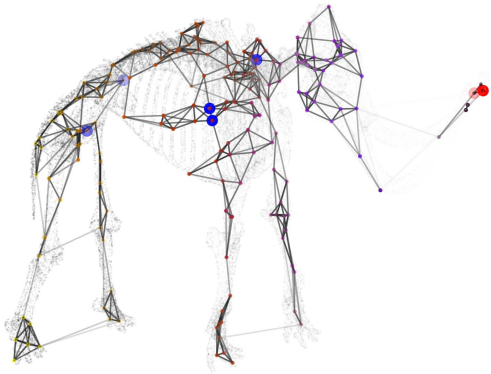
# Fitting low dimensional representation

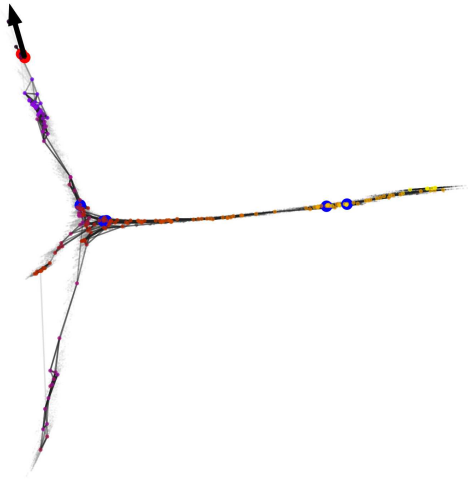
Cross-entropy loss counting positive and negative atraction forces

$$C_{UMAP} = \sum_{i \neq j} v_{ij} \log\left(\frac{v_{ij}}{w_{ij}}\right) + (1 - v_{ij}) \log\left(\frac{1 - v_{ij}}{1 - w_{ij}}\right) \quad (1)$$

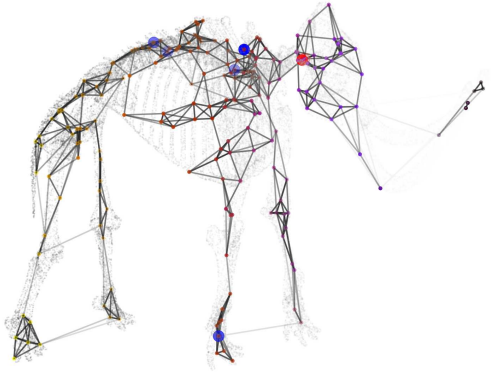
Optimization with **Stochastic Gradient Descent**

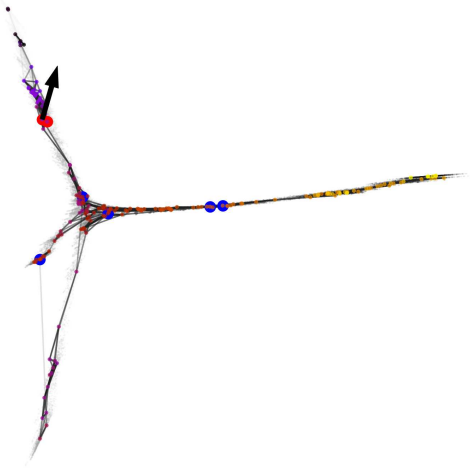
# Fitting low dimensional representation





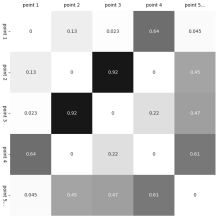
# Fitting low dimensional representation

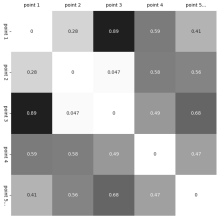


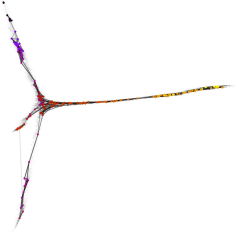


# Fitting low dimensional representation

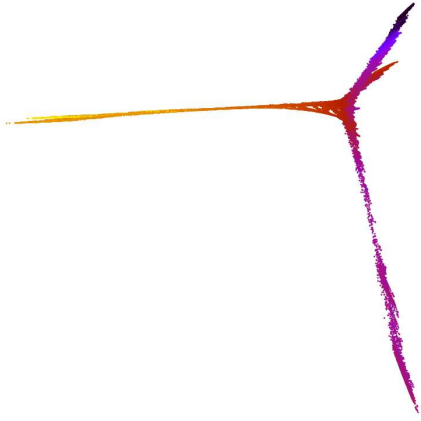




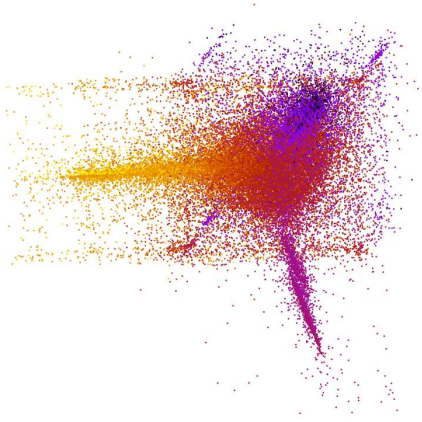



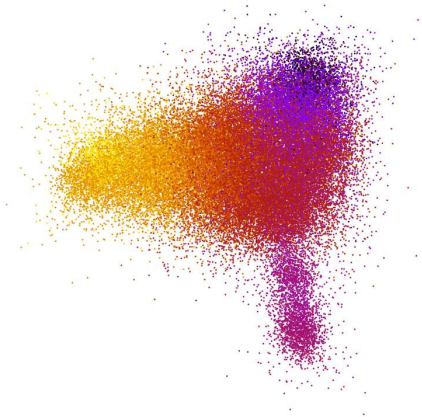
# Optimization



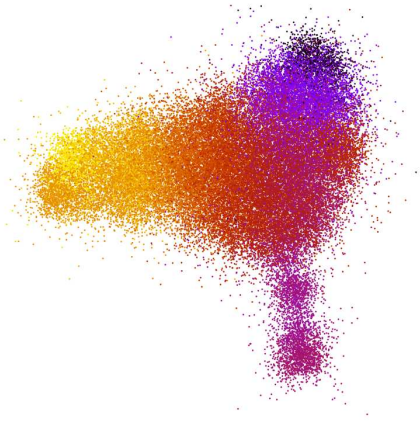
# Optimization



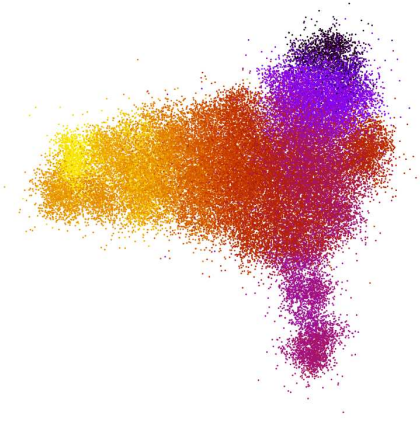
# Optimization



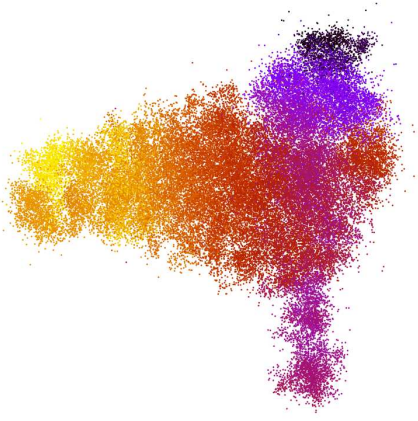
# Optimization



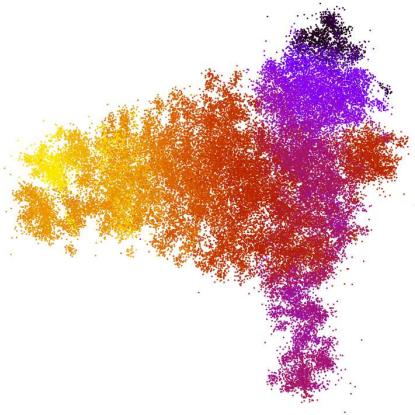
# Optimization



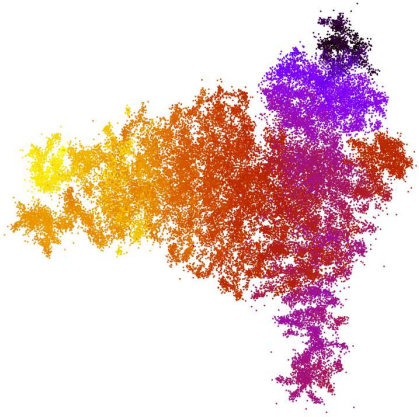
# Optimization



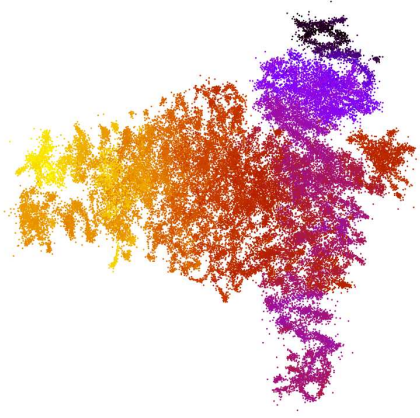
# Optimization



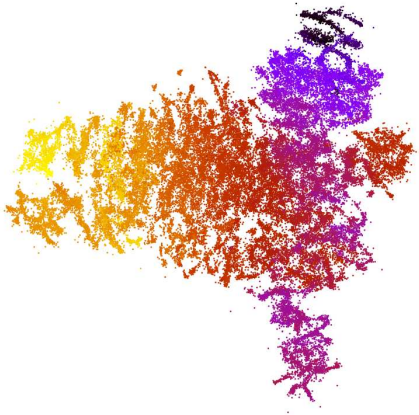
# Optimization



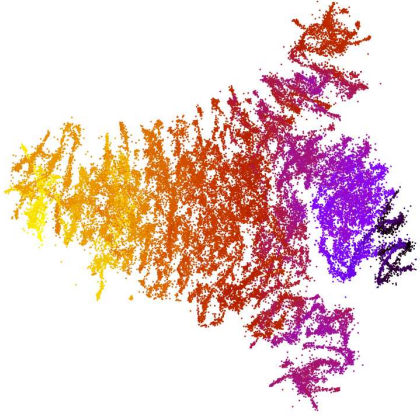
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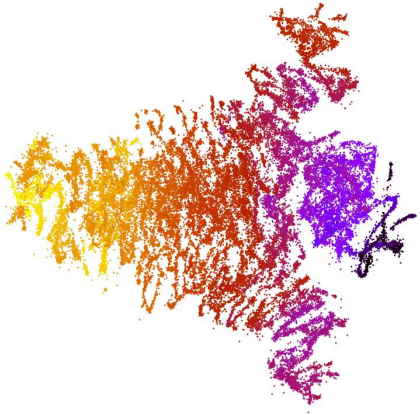
# Optimization



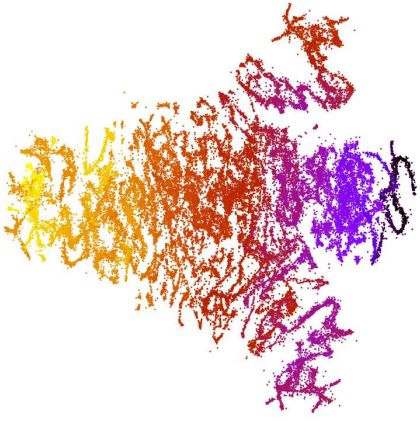
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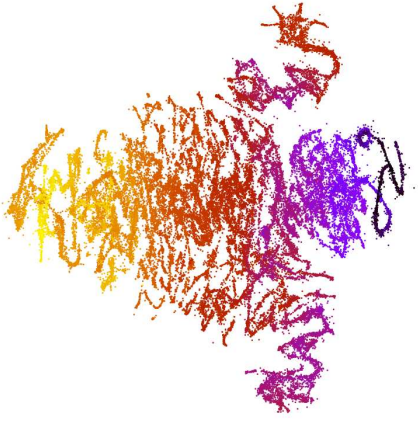
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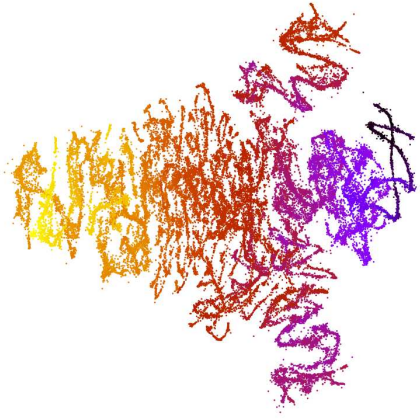
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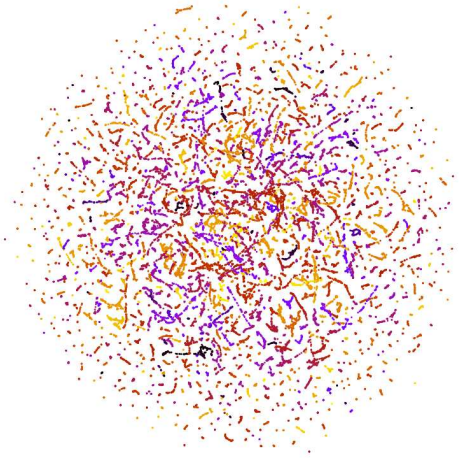
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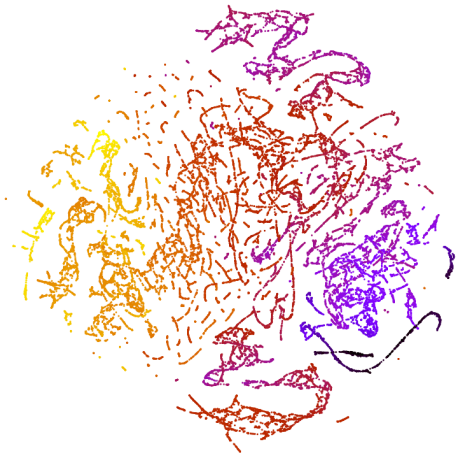
# Basic parameters

| Parameter | Meaning | Default value |
|---|---|---|
| `n_components` | target dimensions | `2` |
| `n_neighbors` | influence of neighborhood | `15` |
| `min_dist` | how points are packed | `0.1` |
| `metric` | how distance is measured | `euclidian` |

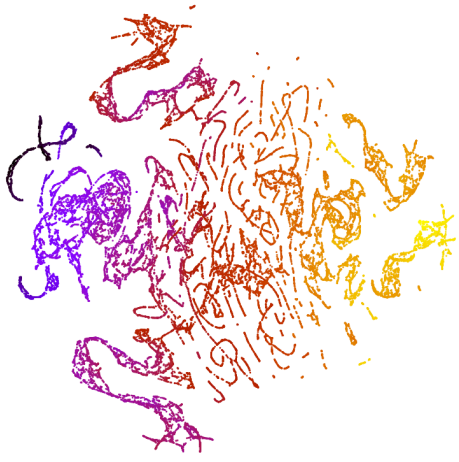
# Influence of the number of neighbors



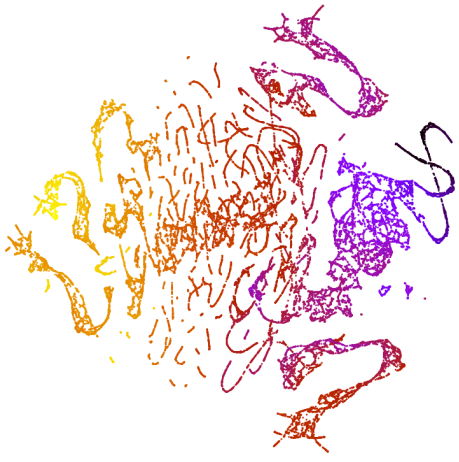
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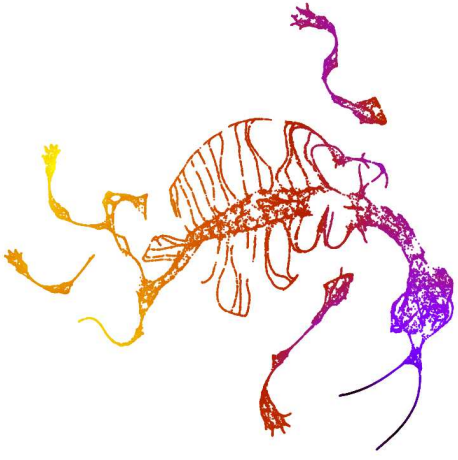
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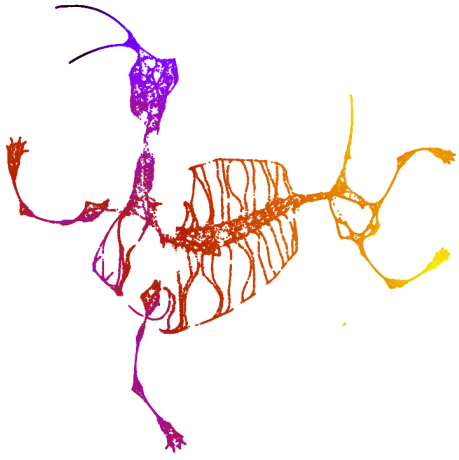
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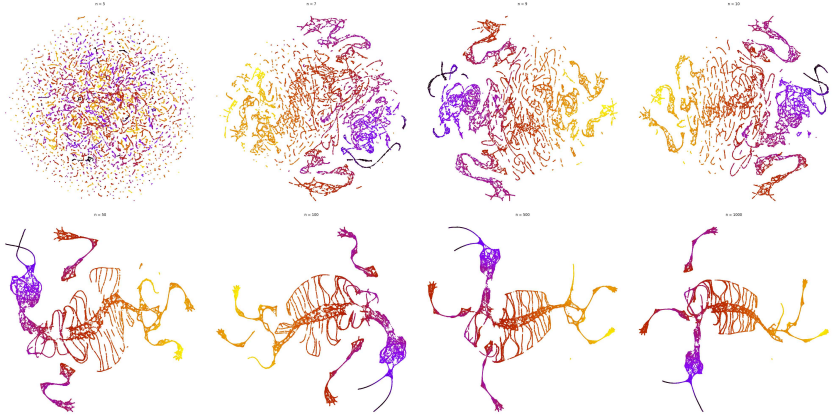
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# Influence of the number of neighbors



# Influence of the number of neighbors



# UMAP vs. T-SNE

## UMAP

- graph theory, fuzzy logic

## T-SNE

- probabilities

# UMAP vs. T-SNE

## UMAP

- graph theory, fuzzy logic
- determinist initialization

## T-SNE

- probabilities
- random initialization

# UMAP vs. T-SNE

## UMAP

- graph theory, fuzzy logic
- determinist initialization
- $log_2$ similarity scores

## T-SNE

- probabilities
- random initialization
- gaussian distribution similarity

# UMAP vs. T-SNE

## UMAP

- graph theory, fuzzy logic
- determinist initialization
- $log_2$ similarity scores
- update by pairs

$\rightarrow$ scales well for large datasets

## T-SNE

- probabilities
- random initialization
- gaussian distribution similarity
- update all points

scales less

# Scaling

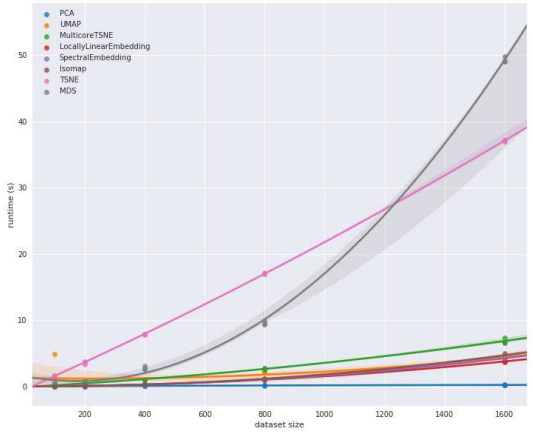



Figure from umap-learn documentation

# Possible traps

- Hyperparameters matter (see `n_neighbors` examples)

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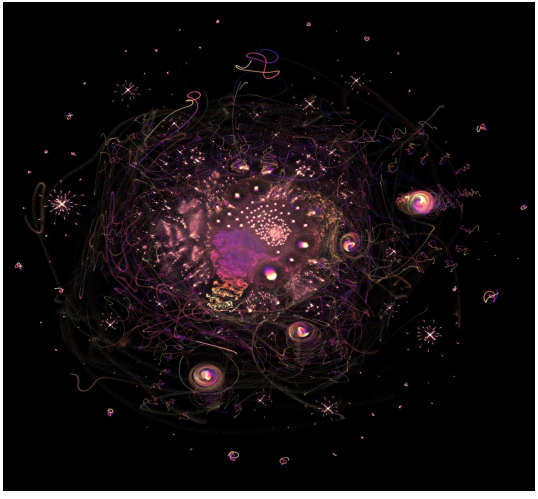
# Possible traps

- Hyperparameters matter (see `n_neighbors` examples)
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# Possible traps

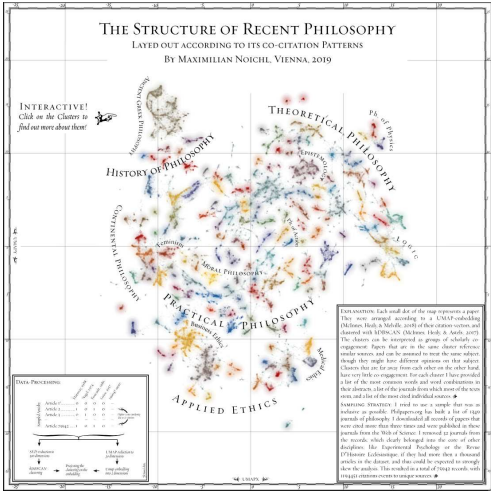
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- Cluster sizes mean nothing (uniformization)
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- Algorithm is stochastic

# Cool examples



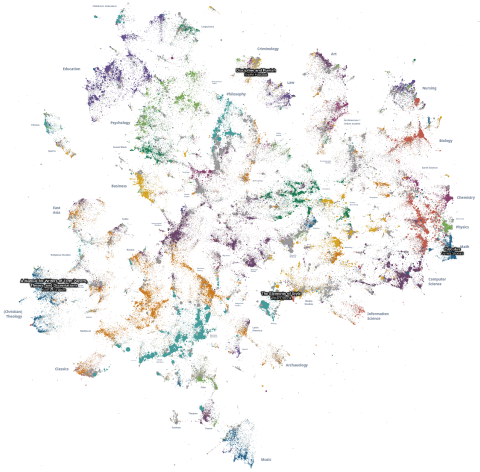
1 million integers by divisibility

# Cool examples




Structure of Recent Philosophy

# Cool examples




Open Syllabus Galaxy

# Cool examples

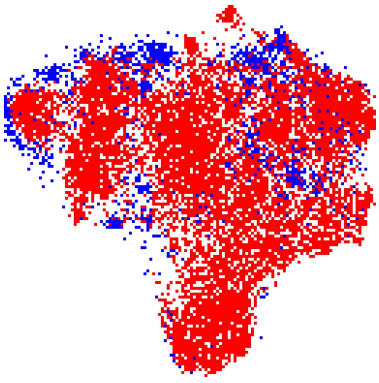






H. Goëau - TDWG 2021

# Cool examples

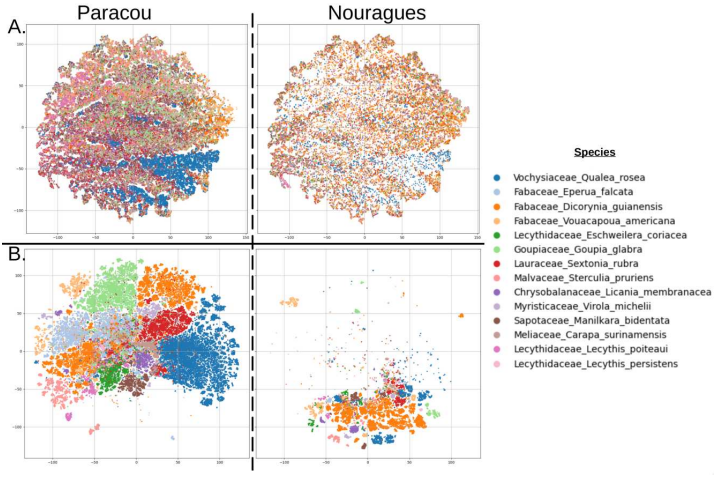






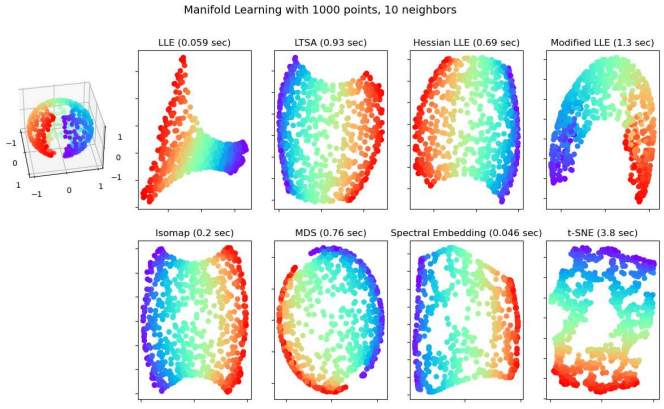
H. Goëau - TDWG 2021

# Cool examples




C. Prieur PhD thesis

# A lot of options !



# Useful ressources

- `umap-learn`
- `scikit-learn`
- Topos institute - understanding UMAP
- Google PAIR - understanding UMAP

on YouTube:

- Talk by Leland McInnes
- deepia
- StatQuest
- DeepFindr

# Thanks for your attention !

Let’s practice !

# References i


Du, Trina Y (2019). **“Dimensionality reduction techniques for visualizing morphometric data: Comparing principal component analysis to nonlinear methods”.** In: *Evolutionary Biology* 46.1, pp. 106–121.

Hinton, Geoffrey E and Sam Roweis (2002). **“Stochastic neighbor embedding”.** In: *Advances in neural information processing systems* 15.

McInnes, Leland, John Healy, and James Melville (2018). **“Umap: Uniform manifold approximation and projection for dimension reduction”.** In: *arXiv preprint arXiv:1802.03426*.

Van der Maaten, Laurens and Geoffrey Hinton (2008). **“Visualizing data using t-SNE.”.** In: *Journal of machine learning research* 9.11.